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Assignment Research: Report 1 NMAP

What did you do?

For this report, I analyzed an organization's network using Nmap version 7.79, an open-source tool for network discovery and security auditing. I used the Nmap-Zenmap GUI instead of just using the CLI because the GUI's tabbed layout makes the report easier for new users like me to understand. I also used the Zenmap GUI User's Guide to help analyze and interpret the results.

I received permission from a local church to perform the scan, believing that many churches may be unaware of potential network security risks. Under the supervision of the church's tech support, I installed the software on an office computer. We reviewed the topology tab, host list, and icon indicators shown in the GUI. Later, I was sent a copy of the scan output and the topology map for further analysis.

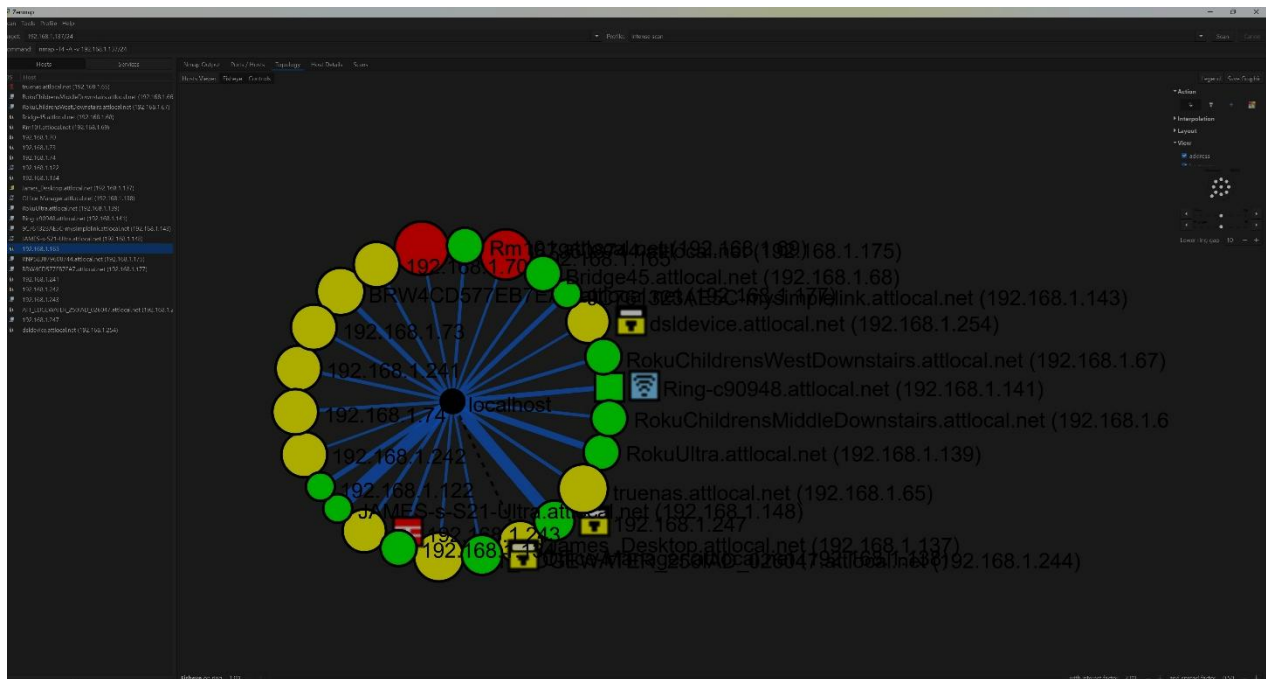
I ran an intense scan across the subnet to identify live hosts. The GUI command executed was:

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nmap -T4 -A -v [IP/Subnet]
```

This command included Pre-Scan and Host Discovery, DNS Resolution, Port Scanning, Service and OS Detection, and NSE Script execution. I saved both a PDF of the output and an image of the topology map.

What are the results?

The scan output was a 26-page-long PDF that will be included with this report. Along with the screenshot of the topology tab from the Zenmap GUI below.



The topology tab shows that there are 25 hosts:

- 12 that have fewer than 3 open ports.
- 11 that have 3-6 open ports.
- 2 that have more than 6 open ports.

The map also showed:

- A firewall on the Edgewater Ethernet switch.
- Two hosts with some filtered ports (an iPhone and a Dell computer).
- A Ring doorbell acting as a wireless access point (WAP).

The highest traceroute RTTs were for the iPhone at address 192.168.***.*** and the device at 192.168.***.***. While not all hosts were labeled with accurate icons, the additional tabs in Zenmap helped identify them.

Note: Nmap doesn't directly support Bluetooth scans, so I couldn't check if Bluetooth was enabled on devices like the Samsung smartphone.

Detected Devices:

- Storage and Servers: TrueNAS server running FreeBSD with 6 open ports.
- IoT Devices: Security cameras, Ring doorbell.
- Printers: Ricoh (The organization's main printer) and a Brother printer that was in the office where the network scan was performed.
- Streaming devices: Roku devices used open port 7000 (AirTunes RTSP).
- Wireless Access Points: Ruckus and Ubiquiti devices with web interfaces, SSH access, and SSL certificates.
- Mobile devices: Samsung S21 smartphone, iPhone.

Protocols Observed:

- Common Services: HTTP/HTTPS (ports 80/443), FTP (21), Telnet (23), SSH (22), SMB (139/445), RTSP (554), and printing protocols (IPP 631, JetDirect 9100).
- IoT/Multimedia Protocols: RTSP for media streaming, SOAP (8082), and gSOAP for device APIs.
- Security protocols: SSL/TLS (some using outdated certificates), and SMB signing that is sometimes optional.

Attack Surface Observations:

- Insecure Services: Telnet, FTP, and SMBv1 expose credentials and traffic in plaintext. These are unencrypted and vulnerable to interception.
- Default Credentials/Anonymous Access: Some devices allowed anonymous FTP access or exposed HTTP admin pages without requiring credentials.
- IoT Exposure: Devices like cameras and smart home devices with minimal security increase risk.
- Web Interfaces: Many devices expose web servers on ports 80/443 or 8082, with some not requiring authentication.
- Outdated Software: Detected firmware and SSL certificates suggest unpatched devices or expired configurations.
- Cloud: Devices such as Ring and myQ possibly communicate with cloud services, but no segmentation was noted.
- Wireless Risks: Android devices and wireless apps could be vulnerable without segmentation or strong authentication. Ruckus WAPs were identified but need further evaluation for WPA configurations, guest network isolation, and management access.

These weaknesses—if unmanaged—could be exploited through misconfiguration, outdated software, or insecure protocols.

What did you learn?

From this assignment and the preparation for this assignment, I have learned a lot! I learned that digital networks are the sum of all the devices connected to communicate and transmit data. I learned how to use Nmap to map a network (Why have I never done this before?). The way that nmap works is by sending packets to a target system and interpreting the responses. The responses reveal information about the system used to create a map of the

network. The Nmap-Zenmap GUI helped me to visualize everything on a network using the topology tab. The Host Details tab helped me to understand which exact devices were mapped out due to all the details provided. The OS icons and vulnerability icon indicators enhanced my ability to interpret scan results. I was able to see every port for a specific host under the Ports/Hosts tab and how those changed by clicking on a different Host in the left side panel. I learned more about the devices connected to a network in depth using the Zenmap GUI instead of just looking at the Nmap that a CLI would have given me, also.

Mapping tools like Nmap-Zenmap can provide a detailed picture of all active devices, services, and protocols in a network – even the overlooked ones like printers or a Google Home. I mapped my home network, and it took me a while to realize that the “Samsung” listed was my smart TV. This exercise made me more aware that the attack surface is larger than it seems. Every network device is a potential entry point. Devices like the Ricoh printer with ten open ports can become risks. Seeing Telnet in the scan results, despite its known vulnerabilities, was particularly concerning—especially since the scan showed that a required password was not set.

I’ve learned that IoT and mobile devices can expand the attack surface, even though they may not seem like something of significance. However, with things like the camera doorbell, others can gain access and sometimes view the private video feed. This is similar to the San Diego Cybersecurity Case we read over for class. Regardless of network size, routine scans can help identify unexpected or vulnerable devices. For example, I noticed a dormant device on my home network and realized other frequently used devices weren’t showing up, prompting me to investigate further.

Ways that this data can be of value to the organization in the future. One is that this mapping can be used to create a baseline of devices and services. Another way is to regularly

scan and analyze networks to reduce risks. On the business side, it's good to make sure devices are consistently updated to eliminate known vulnerabilities quickly. This assignment has helped me learn that it's important to patch! Lastly, mapping can also help to plan for security improvements and vulnerability management.

This assignment highlighted how vital it is to proactively monitor and secure every connected device. Regular mapping and analysis can prevent small issues from becoming big threats.